

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.2: Reflection — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.2: Reflection
Driving Question	DRIVING QUESTION: Why does a mirror show a perfect copy of the world — and where exactly does that copy live?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Symmetry Safari: Lines of Symmetry in Plane Figures	A bold letter 'A' placed beside a plane mirror in the classroom produced a mirror image that appeared laterally reversed (flipped left-right). Leaves from the school compound, paper cut-outs of rectangles, squares, equilateral triangles, circles, and Kanga fabric swatches were folded and tested with a plane mirror. Different shapes yielded different numbers of lines of symmetry: circle (infinite), square (4), rectangle (2), equilateral triangle (3), scalene triangle (0).	A line of symmetry divides a plane figure into two congruent halves that are exact mirror images of each other. It is also called a mirror line. Shapes can have zero, one, more than one, or infinitely many lines of symmetry depending on their geometry. The fold test and the mirror test both confirm the same lines of symmetry, showing that folding and reflecting are geometrically equivalent operations.	The mirror line acts as the line of symmetry — it is the invisible geometric rule that governs where the mirror image appears. When the letter 'A' is placed beside a mirror, the vertical line of symmetry of the letter aligns with the mirror line, producing a reversed image that is congruent to the original. The 'flip' observed in Aisha's bedroom mirror and in the matatu's rear-view mirror is a consequence of reflection across a mirror line, which reverses the figure laterally while preserving all distances and sizes. This is the foundational property that the rest of the unit will make precise and quantitative.
Lesson 2 Mirror, Mirror on the Wall — Properties of Reflection on a Plane Surface	Learners stood at measured distances (20 cm, 40 cm, 60 cm, 80 cm) from a plane mirror and recorded the apparent position of their image. They then constructed image points for four object points on plain paper using a ruler and set square, measuring object distance, image distance, and the angle the joining line made with the mirror line for each point. They also compared the sides of an object triangle with those of its image triangle.	From their measurements learners discovered and formalised three properties of reflection on a plane surface: (1) object distance from the mirror line equals image distance from the mirror line; (2) the line joining an object point to its image is perpendicular to the mirror line; (3) the image is congruent to the object. They applied these to calculate Otieno's total distance from his image (2.4 m) in a Kenyan context and annotated Aisha's	The three formalised properties explain why Aisha's bedroom mirror shows her image the same distance behind the glass as she stands in front (Property 1), why the image appears directly 'across' the mirror rather than at an angle (Property 2), and why every detail — the Kanga pattern, the button count, the letter A — is preserved in size and shape even though it is laterally reversed (Property 3). The matatu driver's rear-view mirror follows the same three

		bedroom mirror scenario with all three properties.	rules, so vehicles behind appear at a true proportional distance.
Lesson 3 Drawing Images on a Plane Surface — Construction Practice	Learners watched the teacher construct the reflected image of a triangle and a rectangle across vertical and horizontal mirror lines, narrating each geometric step (perpendicular from vertex to mirror line; measuring and marking equal distance on the opposite side; joining image vertices). They then observed their own constructions take shape on plain paper using rulers and set squares.	The image of any point P under reflection lies on the perpendicular from P to the mirror line m, at a distance equal to the distance from P to m — on the opposite side. This means m is the perpendicular bisector of the segment joining any object point to its image. This rule holds regardless of whether m is horizontal, vertical, or oblique, and explains precisely why Aisha's image appears exactly as far behind her bedroom mirror as she stands in front of it.	Using the perpendicular-bisector construction rule, learners can now explain: (1) why the image is the same size as the object (congruence is preserved because every vertex is mapped by the same equal-distance rule); (2) why the image appears 'behind' the mirror at the same distance as the object is 'in front' ($d_o = d_i$ by construction); and (3) why the image is laterally reversed (the perpendicular mapping flips the left–right orientation). These three facts together answer the core puzzle of the phenomenon — the driving question 'where exactly does the copy live?' now has a geometric construction answer.
Lesson 4 Reflection on the Cartesian Plane I — Mirror Lines Along the Axes ($x = 0$ and $y = 0$)	When triangle and rectangle vertices in Quadrant I were reflected in the y-axis ($x = 0$), only the x-coordinate changed sign while y stayed the same; when reflected in the x-axis ($y = 0$), only the y-coordinate changed sign while x stayed the same — confirmed across all eight vertices of the Nairobi school-grounds rectangle.	Reflection in the y-axis follows the rule $(x, y) \rightarrow (-x, y)$; reflection in the x-axis follows the rule $(x, y) \rightarrow (x, -y)$. In both cases the mirror line is the perpendicular bisector of the segment joining each object point to its image, consistent with the three properties established in Lesson 2.	The coordinate rules explain precisely why Aisha's image appears laterally reversed: standing at $x = 3$ in front of her mirror wall (the y-axis, $x = 0$), her image is located at $x = -3$, so the left–right (x) direction is flipped while the up–down (y) direction is preserved — which is exactly why the image raises the opposite hand. The 'invisible rule' governing where the image lands is now a coordinate transformation.
Lesson 5 Reflection on the Cartesian Plane II — The $y = x$ Mirror Line	Learners plotted three object points (A(2,5), B(-1,3), C(4,-2)) on a Cartesian grid and used a ruler to construct their reflections in the line $y = x$, recording both the image coordinates and the perpendicular distances from each point to the mirror line.	Learners induced and stated the rule that reflection in the line $y = x$ maps every point (x, y) to the image point (y, x) — the two coordinates swap; they verified that the mirror line is the perpendicular bisector of the segment joining object to image; and they completed a three-row summary table comparing the transformation rules for all three Cartesian mirror lines studied ($x = 0$, $y = 0$, $y = x$).	Learners explained that a diagonal mirror at 45° (the line $y = x$) swaps the two coordinates of a point, in contrast to the y-axis mirror (which negates the x-coordinate) and the x-axis mirror (which negates the y-coordinate); they connected this to the anchoring phenomenon by articulating that any mirror line — whether Aisha's physical bedroom mirror, the matatu side mirror, or a line on the Cartesian plane — imposes a precise, predictable mathematical rule that determines exactly where the image copy lives.
Lesson 6 Finding the Mirror Line: Reverse Reflection on Plane Surfaces and the	Learners observed that when two congruent, laterally-flipped figures are drawn with no mirror line, the perpendicular bisectors of segments joining corresponding	Learners learned that the mirror line is the perpendicular bisector of the segment joining any object point to its corresponding image point; that this property holds	Learners explained why the mirror line must be both the midpoint locus AND perpendicular to each object–image segment — it is the set of all points

Cartesian Plane	points all coincide at a single line — the mirror line — on both plain paper (via compass construction) and the Cartesian plane (via midpoint and perpendicular equation). They also observed that this reverse-engineering method can determine the exact placement of a junction mirror given the positions of a driver and a road hazard.	universally on plain surfaces and the Cartesian plane; that the mirror line can be constructed geometrically using compass arcs or determined algebraically by finding the midpoint and the equation of the perpendicular line; and that this mathematical rule underpins the real-world positioning of road-safety mirrors by traffic engineers in Kenya.	equidistant from the object and its image, meaning any other line through the midpoint would produce a different, non-congruent image. They connected this reasoning to the anchoring phenomenon (the Nairobi bedroom mirror and matatu side mirrors) and to the junction-mirror engineering context, articulating that the 'invisible rule' governing where the reflection lives is the perpendicular bisector relationship, discoverable in both directions of the reflection task.
Lesson 7 Real-Life Applications of Reflection — Group Investigation and Peer Presentations	Learners observed — through group research, sketching, and peer presentations — that driving mirrors on Kenyan matatus show an image of the vehicle behind at the same apparent distance as the real vehicle is from the mirror; that Kanga and kitenge textile patterns are constructed by reflecting a base motif across one or more mirror lines, creating bilateral symmetry; and that landmark Kenyan buildings such as the Parliament Buildings in Nairobi and mosques in Mombasa's Old Town have façades with clear vertical lines of symmetry that are geometrically equivalent to mirror lines in the plane.	Every real-life application of reflection — whether a road-safety mirror, a fabric print, or a building façade — works because of the same three geometric properties established in Lesson 2: (1) the image is as far behind the mirror line as the object is in front; (2) the line joining any object point to its image is perpendicular to the mirror line; (3) the image is congruent to the object. The mirror line is always the perpendicular bisector of the object–image segment. These properties are invariant across all contexts, regardless of whether the mirror line is physical glass, a fold-line in a fabric, or an axis of architectural symmetry.	These findings directly and completely answer the anchoring phenomenon. Aisha's reflection raises the opposite hand because reflection reverses lateral orientation across the mirror line — a geometric consequence of the perpendicular-bisector rule. Her image appears exactly as far behind the mirror as she stands in front because the mirror line is equidistant from object and image. The matatu driver's rear-view mirror shows vehicles at their true apparent distance for the same reason. The Kanga skirt pattern Aisha wears to school is itself a designed instance of reflection symmetry. Even the Parliament Buildings Aisha passes on the matatu ride display the same geometric principle. The 'invisible rule' governing where the image lands is not invisible at all — it is the perpendicular bisector of the object–image segment, and it governs every mirror in Kenya.
Lesson 8 Model Building, Review and Assessment — The Mirror Fully Explained	Learners observed — through their own Brain Dump, Gap-Spotter checklist, and final model diagram — the full architecture of reflection geometry: lines of symmetry dividing plane figures into congruent halves; the three properties of plane-surface reflection (equal distances, perpendicularity, congruence); Cartesian transformation rules for $x = 0$, $y = 0$, and $y = x$; and the perpendicular-bisector method for locating a hidden mirror line. They also observed, concretely, how much their own models had grown since Lesson 1.	The geometry of reflection is governed by one unifying principle: the mirror line is the perpendicular bisector of the segment joining every object point to its corresponding image point. This single principle explains all three properties (equal distances, perpendicularity, congruence), generates all three Cartesian transformation rules, and makes it possible to reconstruct any mirror line from an object–image pair. Lines of symmetry are the special case where object and image are the same figure.	The anchoring phenomenon is now fully explained. Aisha's image appears the same distance behind her bedroom mirror as she stands in front because the mirror surface is the perpendicular bisector of every object–image segment — equal distances are a mathematical consequence, not a coincidence. The image is the same size because reflection is a congruence transformation. The image is laterally reversed because the perpendicular mapping preserves distances along the mirror but flips the perpendicular direction,

			swapping left and right. The matatu rear-view mirror obeys identical properties. Every real-life mirror — bedroom glass, road-safety mirror, rear-view mirror — is a physical instantiation of the same perpendicular-bisector geometry that learners have now mastered.
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ATTRIBUTION AND LICENSING

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